Dominik Szewczyk

Data Scientist · Machine Learning & AI

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PROFILE

Data Science and AI graduate (cum laude) who takes projects from start to finish. Across two internships and client work, I cover the whole process, from data analysis and machine learning to a working tool that meets the client's needs.

// 01 Experience

Data Scientist, Graduation Internship · KapiteinLabs

Dec 2025 - Jun 2026

- Built a predict-and-optimise machine learning system, delivered as a **no-code Streamlit dashboard** that engineers and clients use without writing code.
- Trained a physics-informed neural network (PINN) to predict ozone output from reactor settings, reaching **R^2 = 0.988, MAE = 1.49 g/kWh, MAPE = 1.9%** on 438 measurements from TU Eindhoven.
- Added an inverse optimiser that recommends the settings needed to hit a target output, replacing slow one-off lab experiments with instant estimates.

Data Scientist, Internship · De Rooy Transport & Logistics

Feb - Jun 2025

- Built a **multi-model pipeline** forecasting where, when, and how large unexpected pickups will be (over 82% of daily assignments), wired into the planners' PTV route software.
- Pickup occurrence: a Keras model over 5 km grid cells reaching **90.2% accuracy** and 86.5% F1; location via random forest with an **18 m mean error** and 100% of predictions within 1 km.
- Engineered spatial, temporal, and holiday features for peak robustness. Impact: up to **€1,500, 1,500 km, and 945 kg CO_2 saved per day**. Featured in the LSB whitepaper.

Web Developer · Freelance

Feb 2026 - Present

Built and deployed 10 live websites for small-business clients, owning each project from design through to deployment.

// 02 Selected Projects

AI 3D Digital Twin & Object Removal

Antea Group & Kavel 10 · 2024 to 2025

Reconstructed urban environments from aerial and drone imagery. Benchmarked DeepFill v2, LaMa, and PowerPaint for object removal and U-Net, DeepLabV3, and FCN for segmentation, then built 3D meshes with Instant NeRF and a VR walkthrough in Unreal Engine.

Computer Vision · NeRF · Semantic Segmentation · 3D Reconstruction

Plant Analysis with Computer Vision & Robotics

NPEC · 2023 to 2024

Segmented seeds, shoots, and roots from plant images with a U-Net and instance segmentation, localising root tips to within roughly 1 mm. Simulated a PID-controlled liquid-handling robot and applied reinforcement learning to dispense precisely at the root tips.

Computer Vision · U-Net · Reinforcement Learning

ATHENA: MLOps for Plant Science

NPEC · 2024

Took ATHENA research models to production as a Python package and cloud app for organ segmentation and landmark detection. Built Azure pipelines, containerised with Docker, and served via FastAPI; recognised publicly by NPEC.

Microsoft Azure · Docker · FastAPI · MLOps

Emotion Classification with NLP

Banijay Media · 2024

Detected emotions in video content by classifying text extracted from clip fragments. Compared Naive Bayes, logistic regression, HMM, CRF, RNN, and LSTM, then fine-tuned BERT to reach an F1 above 0.88 on the held-out test set.

NLP · Transformers (BERT) · Feature Engineering

// 03 Education

Breda University of Applied Sciences · Data Science & Artificial Intelligence

2022 - 2026

Graduated cum laude. Two industry internships and applied research in ML, computer vision, NLP, and MLOps.

II LO im. Andrzeja Frycza Modrzewskiego · Extended Mathematics

2019 - 2023

One of the top-ranked secondary schools in the region.

// 04 Skills & Tools

MACHINE LEARNING & AI	Machine Learning, Deep Learning, Computer Vision, NLP & Transformers, Predictive Modeling
LANGUAGES & TOOLS	Python, scikit-learn, TensorFlow, Keras, XGBoost, FastAPI, Streamlit, Power BI, R, Git
CLOUD & MLOPS	Microsoft Azure, Azure ML, Docker, CI/CD pipelines
WAYS OF WORKING	Agile (Scrum), data visualisation, stakeholder communication
LANGUAGES	English (fluent) · Polish (native) · German (basic)
INTERESTS	Football, gym, chess (3rd category), half marathon (1:43)